UNIVERSITEIT VAN PRETORIA | UNIVERSITY OF PRETORIA

Departement Wiskunde en Toegepaste Wiskunde | Department of Mathematics and Applied Maths

11 Maart / March 2002
Punte / Marks: **35**
Tyd / Time: **60** min

Punte Marks	28

WTW 158: Calculus
SEMESTERTOETS 1 / SEMESTER TEST 1

Van / Surname: MAMMEN

VOORNAME / FIRST NAMES: SIJU ABRAHAM

STUDENTNO: 2 2 0 2 0 5 8 7 HANDTEKENING /SIGNATURE: Siju Mammen

STUDIERIGTING/FIELD OF STUDY B. ENG. (COMPUTERS)

Omkring jou lesinggroepnommer op die tabel / Encircle your lecture group number on the table

Lesinggroepnommer Lecture group number	Dosent Lecturer	Lokaal Venue	Taal Language	Periode Period
1	Mev / Mrs AE du Preez	GW 4-3	Afrikaans	B
2	Mev / Mrs L Mostert	Ing II 3-40	Afrikaans	B
3	Prof MJ Schoeman	Louw	English	B
4	Prof MJ Schoeman	GW 4-3 & GW 4-1	Afrikaans	E
5	Mev / Mrs KH Jordaan	Maths 1-1	English	E

Lees eers die volgende instruksies

1. *Slegs nie-programmeeerbare sakrekenaars mag gebruik word.*
2. *Die vraestel bestaan uit bladsye 1 tot 7(vrae 1-11). Kontroleer of u vraestel volledig is.*
3. *Doen alle krapwerk op die blanko bladsy(e). Dit word nie nagesien nie.*
4. *As u meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan ook die blanko bladsy(e) en dui dit duidelik aan deur 'n* **raam daarom te trek.**
5. *Geen potloodwerk word nagesien nie.*
6. *Korreksievloeistof (bv.Tipp-Ex) mag nie gebruik word nie.*
7. *Geen antwoordboek mag uit die lokaal geneem word nie.*
8. *Enige navrae oor die nasienwerk moet binne* **drie dae** *nadat die toetse terugbesorg is, gedoen word. Daarna word aanvaar dat alles korrek is.*
9. ***Toon alle bewerkings duidelik.***

First read the following instructions

1. *Only non-programmable calculators may be used.*
2. *The question paper consists of pages 1 to 7(questions 1-11). Check if your paper is complete.*
3. *Scribbling can be done on the empty page(s). This will not be marked.*
4. *If you need more than the available space for an answer, you may also use the empty page(s). Indicate it clearly by* **framing** *it.*
5. *Work in pencil will not be marked.*
6. *Use of correction fluid (e.g.Tipp-Ex) is not allowed.*
7. *No test scripts may be removed from the venues.*
8. *Any queries about the marking must be done within* **three days** *after the tests have been handed back. After that we assume that everything is in order.*
9. ***Show all computations clearly,***

VRAAG 1 / *QUESTION 1*

A, B en C is die hoekpunte van $\triangle ABC$. Vind $\overline{AB}+\overline{BC}+\overline{CA}$.

A, B and C are the vertices of $\triangle ABC$. Find $\overline{AB}+\overline{BC}+\overline{CA}$.

$$\bar{AB} + \bar{BC} + \bar{CA} = \bar{0}$$ ✓

2

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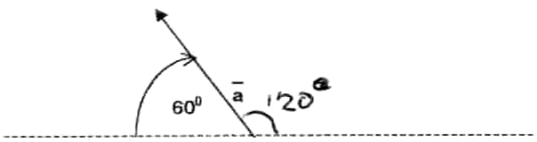
VRAAG 2 / *QUESTION 2*

$\bar{a} \in \mathbb{R}^2$ het lengte 5. As $\bar{a} = a_1\bar{i} + a_2\bar{j}$, bepaal a_1 en a_2.

$\bar{a} \in \mathbb{R}^2$ has magnitude 5. If $\bar{a} = a_1\bar{i} + a_2\bar{j}$, determine a_1 and a_2.

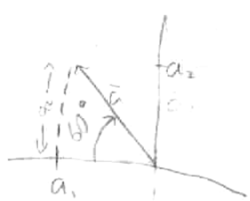


$\bar{a} = \langle a_1, a_2 \rangle$

$|\bar{a}| = 5$

$a_1^2 + a_2^2 = |\bar{a}|^2$

$a_1^2 + a_2^2 = 25$



$\frac{a_2}{a_1} = \cancel{\tan 60}\ \tan 120 = -\tan 60$

$a_2 = -a_1 \tan 60$

$a_1^2 + a_2^2 = 25$ ✓

$a_2 = 4,33$

$a_1^2 + (a_1 \tan 60)^2 = 25$

$a_1^2(1 + \tan^2 60) = 25$

$a_1^2 = \frac{25}{1+\tan^2 60}$

$a_1 = \pm 2,5$ ✓

a_1 in second quad $\rightarrow \therefore a_1 = -2,5$

2

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VRAAG 3 / *QUESTION 3*

Bepaal x (indien moontlik) sodat $P(-1,1,2)$, $Q(1,x,3)$ en $R(2,-1,4)$ op 'n reguit lyn lê.

Find x (if possible) such that $P(-1,1,2)$, $Q(1,x,3)$ and $R(2,-1,4)$ lie on a straight line.

$\bar{PQ} = \langle 1+1;\ x-1;\ 3-2 \rangle$

$= \langle 2;\ x-1;\ 1 \rangle$

$\bar{QR} = \langle 2-1; -1-x; 4-3 \rangle$

$\bar{QR} = \langle 1; -1-x; 1 \rangle$

$\bar{PQ} \cdot \bar{QR} = 1$ (parallel vectors)

$2\ (x-1)(x+1) + 1 = 1$

$2 = x^2 + 1$ ✗

$x^2 = 1$

$x = \pm 1$

0

[3]

VRAAG 4 / *QUESTION 4*

Gegee ΔABC met $A(1,0,-1), B(0,2,1)$ en $C(1,2,0)$. Bepaal die grootte van $\hat{B}$.

Given ΔABC *with* $A(1,0,-1), B(0,2,1)$ *and* $C(1,2,0)$. *Find the magnitude of* $\hat{B}$.

$\overline{BA} = \langle 1-0;\ 0-2;\ -1-1 \rangle$
$= \langle 1;\ -2;\ -2 \rangle$

$\overline{BC} = \langle 1-0, 2-2, 0-1 \rangle$
$= \langle 1;\ 0;\ -1 \rangle$

$\dfrac{\overline{BA} \cdot \overline{BC}}{|\overline{BA}| \cdot |\overline{BC}|} = \cos\theta$

$\dfrac{1+0+2}{(\sqrt{1+4+4})(\sqrt{1+0+1})} = \cos\theta$

$\dfrac{3}{\sqrt{9} \cdot \sqrt{2}} = \cos\theta$

$\cos\theta = \dfrac{1}{\sqrt{2}}$

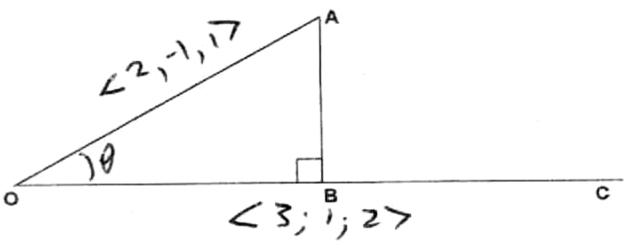
$\theta = 45^\circ$

[3] 3.

VRAAG 5 / *QUESTION 5*

Gegee $\overline{OA} = \langle 2,-1,1 \rangle$ en $\overline{OC} = \langle 3,1,2 \rangle$. Bepaal $\overline{OB}$ en $\overline{BA}$.

Given $\overline{OA} = \langle 2,-1,1 \rangle$ *and* $\overline{OC} = \langle 3,1,2 \rangle$. *Determine* $\overline{OB}$ *and* $\overline{BA}$.



$\overline{OB} = \mathrm{proj}_{\overline{OC}}\, \overline{OA}$

$= \left(\dfrac{\overline{OA} \cdot \overline{OC}}{|\overline{OC}|}\right) \dfrac{\overline{OC}}{|\overline{OC}|}$

$= \dfrac{(\overline{OA} \cdot \overline{OC})\, \overline{OC}}{|\overline{OC}|^2}$

$= \dfrac{(6-1+2)(\overline{OC})}{(\sqrt{3^2+1^2+2^2})^2}$

$= \dfrac{7}{14}\, \overline{OC} = \dfrac{1}{2}\, \overline{OC}$

$\overline{OB} = \left\langle \dfrac{3}{2}, \dfrac{1}{2}, 1 \right\rangle$

$\overline{OB} + \overline{BA} = \overline{OA}$

$\overline{BA} = \overline{OA} - \overline{OB}$

$= \left\langle \dfrac{1}{2}; \dfrac{-3}{2}; 0 \right\rangle$

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VRAAG 6 / *QUESTION 6*

Bepaal 'n eenheidsvektor loodreg op beide $\bar{a} =< 1,-3,2 >$ en $\bar{b} =< -2,1,-5 >$.

Determine a unit vector perpendicular to both $\bar{a} =< 1,-3,2 >$ *and* $\bar{b} =< -2,1,-5 >$.

$$\bar{a}\times\bar{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ -2 & 1 & -5 \end{vmatrix} = \hat{i}(15-2) - \hat{j}(-5-4) + \hat{k}(1+6)$$

$$= \hat{i}\,13 + \hat{j}\,9 + \hat{k}\,7$$

$$= <13; 9; 7>$$

$$|\bar{a}\times\bar{b}| = \sqrt{169+81+49} = \sqrt{299}$$

$$\frac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|} ✓$$

unit vector ~~in~~ ~~those~~ perpendicular to $\bar{a}$ & $\bar{b}$ $= \dfrac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|} = \left\langle \dfrac{13}{\sqrt{299}}, \dfrac{9}{\sqrt{299}}, \dfrac{7}{\sqrt{299}} \right\rangle$

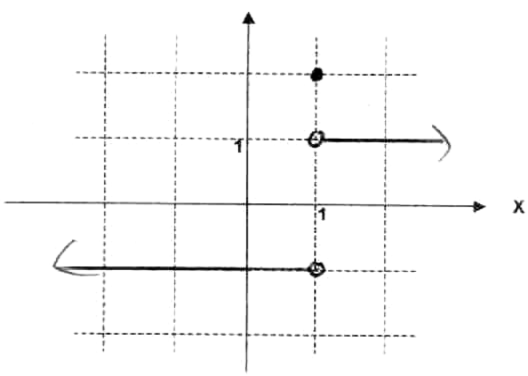
2

[3]

VRAAG 7 / *QUESTION 7*

7.1) Skets / *Sketch* $f(x) = \begin{cases} \dfrac{x-1}{|1-x|} & , x \neq 1 \\ 2 & , x = 1 \end{cases}$

$$= \begin{cases} \dfrac{x-1}{1-x}, & 1-x>0 \\ \dfrac{x-1}{-(1-x)}, & 1-x<0 \\ 2, & x=1 \end{cases}$$



$$= \begin{cases} -1, & x<1 \\ 1, & x>1 \\ 2, & x=1 \end{cases}$$

Gee die waardes van die volgende limiete (as dit bestaan)

Give the values of the following limits (if they exist)

7.1.1) $\lim_{x\to1^+} f(x)$. $= 1$

7.1.2) $\lim_{x\to-1^-} f(x)$. $= -1$

4

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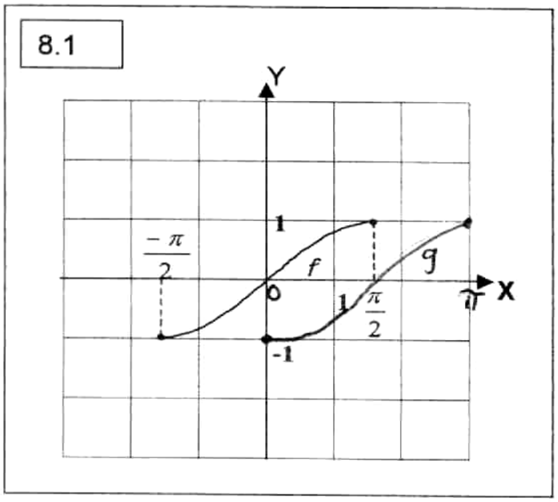
VRAAG 8 / *QUESTION 8*

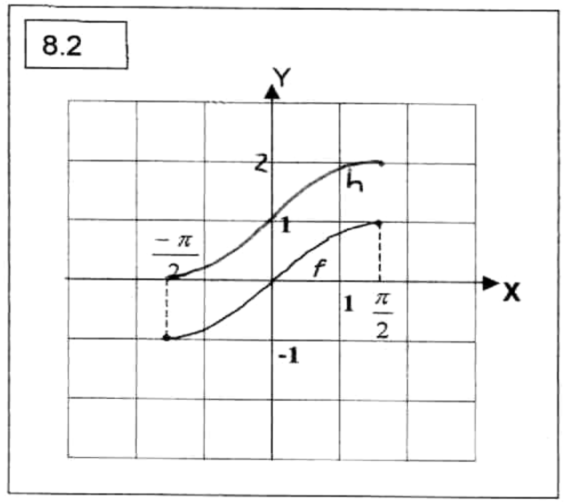
Die grafiek van f is gegee. Op dieselfde assestelsel skets die grafieke van

The graph of f is given. On the same set of axes sketch the graphs of

8.1) $g(x) = f\left(x - \frac{\pi}{2}\right)$

8.2) $h(x) = f(x) + 1.$





8.3) Gee die waarde van $(f^{-1} \circ f)(1)$.

Give the value of $(f^{-1} \circ f)(1)$.

$f(x) = \sin x \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$f^{-1}(x) = \arcsin x \quad -1 \leq x \leq 1$

$(f^{-1} \circ f)(1) = f^{-1}(\sin x)$

$= f^{-1}(\sin 1)$

$= \arcsin(\sin 1)$

$= 1$

3

[3]

$f(x) = \sin x$

VRAAG 9 / *QUESTION 9*

Bepaal die volgende limiete (as dit bestaan)

Evaluate the following limits (if they exist)

9.1) $\lim_{x \to 2} \dfrac{x^2 - 4}{\sin(x-2)}$ $= \lim_{x \to 2} \dfrac{(x+2)(x-2)}{\sin(x-2)}$

$= \lim_{x\to 2}(x+2) \times \lim_{x \to 2}\left(\dfrac{x-2}{\sin(x-2)}\right)$

$= \lim_{x\to 2}(x+2) \times \lim_{x\to 2}\left(\dfrac{\sin(x-2)}{x-2}\right)^{-1}$

$= 4 \times (1)^{-1}$

$= 4$

3

[3]

9.2) $\lim_{x \to 0^+} \left(\dfrac{1}{x} - \dfrac{1}{\sqrt{x}}\right)$ $= \lim_{x\to 0^+}\left(\dfrac{1-\sqrt{x}}{x}\right)$

$= \lim_{x \to 0^+}\left(\dfrac{\frac{1}{x} - \frac{1}{\sqrt{x}}}{}\right)$

$= 0$

$= 0$

2

[2]

1

$\dfrac{1}{x^2} - \dfrac{1}{x}$

$\lim_{x\to 0^+} \dfrac{1}{x} + \lim_{x \to 0^+} -\dfrac{1}{\sqrt{x}}$

$\infty - \infty$

not possible

$\dfrac{x^2 - x}{x^3}$

$\dfrac{x(x-1)}{x \cdot x^2}$

$= \dfrac{x}{x^2} - \dfrac{1}{x^2}$

$= \dfrac{1}{x} - \dfrac{1}{x^2}$

VRAAG 10 / *QUESTION* 10

Gegee dat f kontinu is in $x = 4$ en dat $\lim_{x \to 4} [x - f(x)] = 2$. Vind $f(4)$.

Given that f *is continuous at* $x = 4$ *and that* $\lim_{x \to 4} [x - f(x)] = 2$. *Find* $f(4)$.

$\lim_{x \to 4} [x - f(x)] = 2$

If continuous at $x = 4$, $\lim_{x \to 4} f(x) = f(4)$

consider $f(x)$ to be a polynomial

then $4 - f(4) = 2$

$4 - 2 = f(4)$

$\underline{f(4) = 2}$

2

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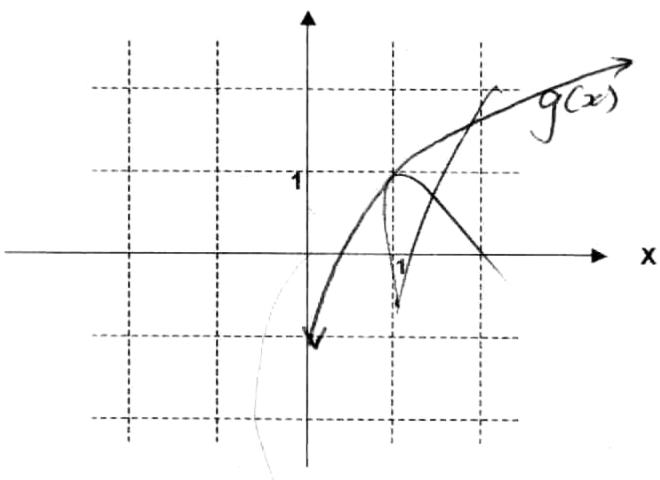
TEGNIEKE / *TECHNIQUES*

VRAAG 11 / *QUESTION* 11

11.1) Skets / *Sketch* $g(x) = \ln(1 + x)$.



11.2) Los op vir x: $\frac{1}{2} \ln x = 1 - \ln 2$. (Sonder 'n sakrekenaar.)

Solve for x: $\frac{1}{2} \ln x = 1 - \ln 2$. (Without a calculator.)

$\frac{1}{2} \ln x + \ln 2 = 1$

$\ln \sqrt{x} + \ln 2 = 1$

$\ln 2\sqrt{x} = 1$

$2\sqrt{x} = e$

$\sqrt{x} = \frac{e}{2}$

$x = \left(\frac{e}{2}\right)^2$

11.3) Bepaal (indien moontlik) $\lim_{x \to 2} \frac{\pi}{2}$.

Evaluate (if possible) $\lim_{x \to 2} \frac{\pi}{2}$. $= \frac{\pi}{2}$

11.4) Laat $f(x) = 2^x$. Gee die vergelyking van g as g simmetries is met f om die lyn $y = x$.
Let $f(x) = 2^x$. Write down or state the equation of g if g is symmetric to f about the line $y = x$.

$g(x) = \log_2 x$

$g(x) = \log_2 x$

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